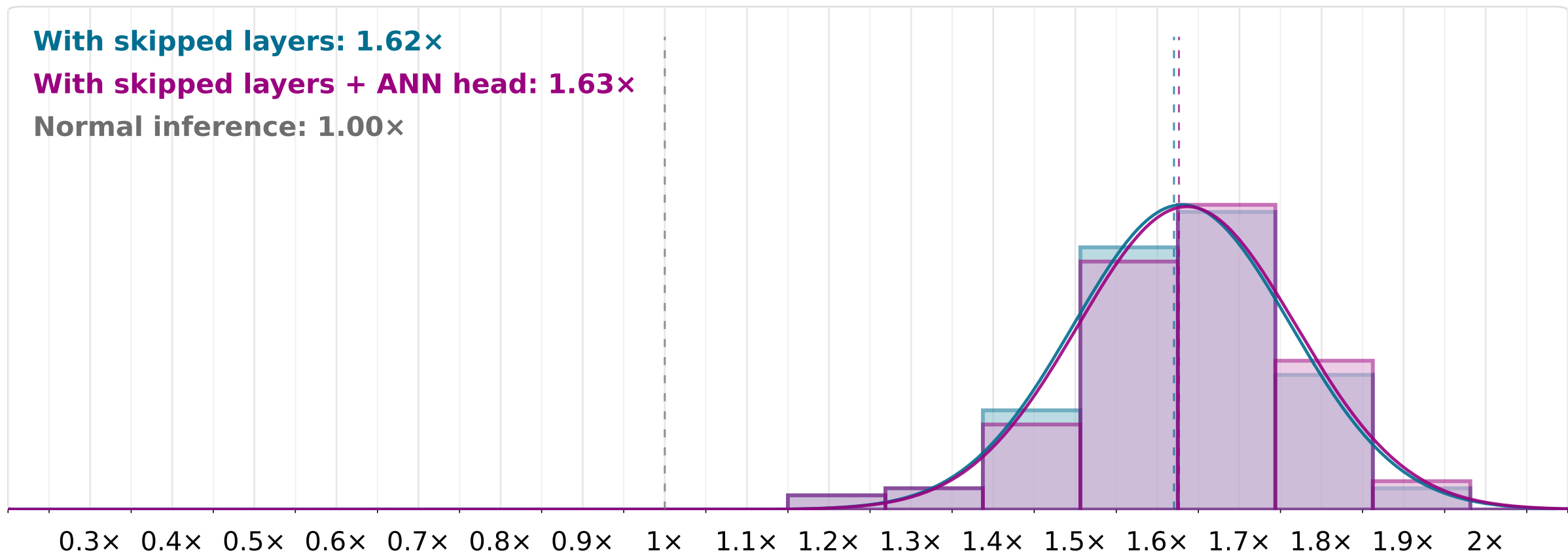


Self-speculation per-token speedup

Mistral 7B Instruct v0.3



Setup

Prompt set = **concrete**
Gap = **(1, 1)**
Block size = **2**
Layers = **32**
LM vocab = **32,768**
LM params = **7.25B**
Head params = **134.2M**
Head portion = **1.85%**

Runtime

GPU = **NVIDIA L4**
Backend = **sdpa**
Model dtype = **bfloat16**
Bridge dtype = **bfloat16**
Speed internals = **no**
Profile prompts = **15**
Warmup prompts = **5**
Measured prompts = **120**

Results

Peak mem normal = **13.60 GiB**
Mean speedup (w/o ANNH) = **1.62x**
Acceptance rate (w/o ANNH) = **51.3%**
Exact match (w/o ANNH) = **71.7%**
Peak mem self (w/o ANNH) = **13.60 GiB**
Mean speedup (w ANNH) = **1.63x**
Acceptance rate (w ANNH) = **50.7%**
Exact match (w ANNH) = **71.7%**
Peak mem self (w ANNH) = **13.64 GiB**
Tokens gen = **s 21,492 / n 21,601**

Profile

Drafter split (w/o ANNH) = **B 75.5% / H 17.7% / O 6.9%**
Verifier/normal (w/o ANNH) = **1.06x**
Drafter/normal (w/o ANNH) = **9.7%**
Drafter split (w ANNH) = **B 85.9% / H 6.3% / O 7.8%**
Verifier/normal (w ANNH) = **1.06x**
Drafter/normal (w ANNH) = **8.5%**
ANNH accuracy = **99.0%**
ANNH head speedup = **3.16x**
ANNH index = **4096 clusters; top-k 100**